

Homework 5

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```
library(ggplot2)
```

```
homicides = read.csv('data/homicide-data.csv') |>  
  dplyr::mutate(city_name = paste0(city, ", ", state))
```

Import data Find the city with the highest homicide and convert to sf

```
homicide.city.count = homicides |>  
  dplyr::group_by(city, state) |>  
  dplyr::summarise(count = dplyr::n()) |>  
  dplyr::ungroup() |>  
  dplyr::arrange(-count)
```

```
head(homicide.city.count)
```

```
## # A tibble: 6 x 3  
##   city      state count  
##   <chr>    <chr> <int>  
## 1 Chicago  IL      5535  
## 2 Philadelphia PA      3037  
## 3 Houston  TX      2942  
## 4 Baltimore MD      2827  
## 5 Detroit  MI      2519  
## 6 Los Angeles CA      2257
```

```
chicago.homicides = homicides |>  
  dplyr::filter(city == 'Chicago',  
                state == 'IL') |>  
  sf::st_as_sf(coords = c("lon", "lat"),  
               crs = 4269) ## crs same as tigris object
```

```
chicago.homicides
```

```
## Simple feature collection with 5535 features and 11 fields  
## Geometry type: POINT  
## Dimension: XY  
## Bounding box: xmin: -87.84657 ymin: 41.64791 xmax: -87.52605 ymax: 42.02271
```

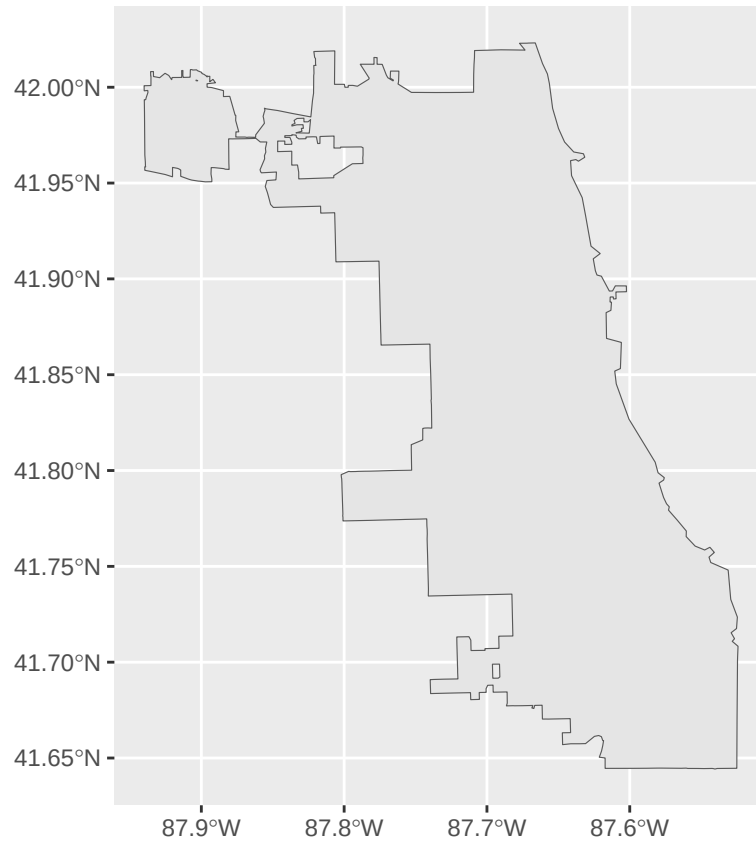
```
## Geodetic CRS: NAD83
## First 10 features:
##      uid reported_date victim_last victim_first victim_race victim_age
## 1 Chi-003284      20070101    THOMAS      LEON        Black        36
## 2 Chi-003285      20070101    ALLEN       MARIO        Black        31
## 3 Chi-003286      20070101    HALL      CHALLAH      Black        21
## 4 Chi-003287      20070101    GUZMAN    CAMERINO    Hispanic     25
## 5 Chi-003288      20070102    ROOT     FRANK       White        34
## 6 Chi-003289      20070104    WHITE    MEREDITH    Black        37
## 7 Chi-003290      20070104    ABUNIMEH  RAMI M      White        31
## 8 Chi-003291      20070108    BELL     DENNIS     Black        38
## 9 Chi-003292      20070111    CAMBELL   RONNIE     Black        29
## 10 Chi-003293     20070112    DENTON    NAJJA      Black        31
##      victim_sex  city state      disposition  city_name
## 1      Male Chicago  IL      Closed by arrest Chicago, IL
## 2      Male Chicago  IL      Closed without arrest Chicago, IL
## 3      Male Chicago  IL      Closed by arrest Chicago, IL
## 4      Male Chicago  IL      Open/No arrest Chicago, IL
## 5      Male Chicago  IL      Closed by arrest Chicago, IL
## 6      Male Chicago  IL      Open/No arrest Chicago, IL
## 7      Male Chicago  IL      Open/No arrest Chicago, IL
## 8      Male Chicago  IL      Open/No arrest Chicago, IL
## 9      Male Chicago  IL      Closed by arrest Chicago, IL
## 10     Male Chicago  IL      Open/No arrest Chicago, IL
##      geometry
## 1 POINT (-87.67493 41.7668)
## 2 POINT (-87.74467 41.89165)
## 3 POINT (-87.58057 41.76514)
## 4 POINT (-87.71251 41.83797)
## 5 POINT (-87.62703 41.85043)
## 6 POINT (-87.73429 41.88064)
## 7 POINT (-87.72888 41.89809)
## 8 POINT (-87.62397 41.79015)
## 9 POINT (-87.66341 41.73853)
## 10 POINT (-87.62472 41.80106)
```

Import chicago vector

```
chicago.boundary = tigris::places(state = "IL", cb = T, class = "sf") |>
  dplyr::filter(NAME == "Chicago")
```

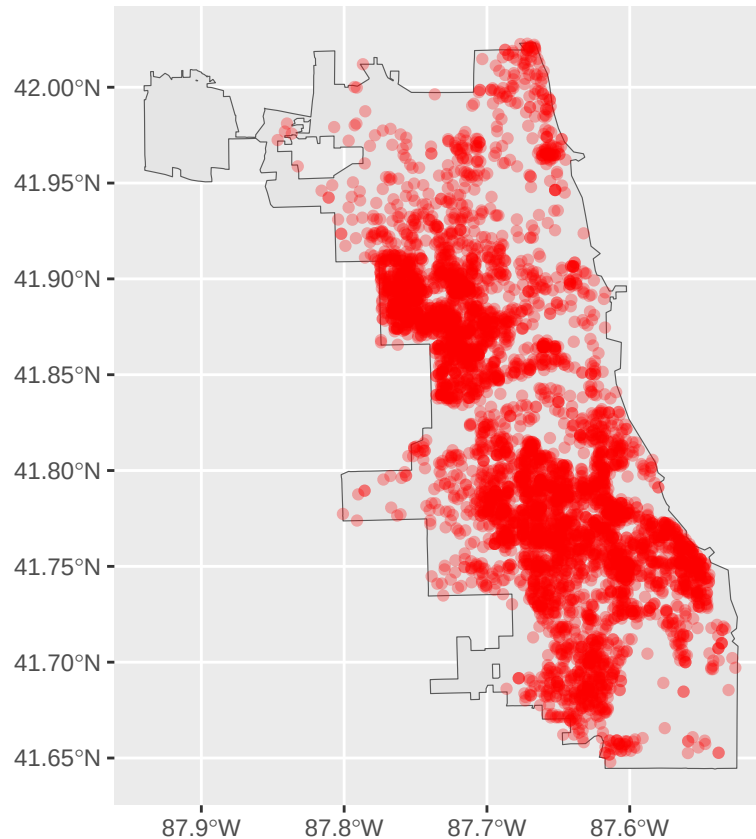
```
## |
```

```
ggplot(data = chicago.boundary) +
  geom_sf()
```



Plot locations of homicides into chicago

```
ggplot() +  
  geom_sf(data = chicago.boundary) +  
  geom_sf(data = chicago.homicides, colour = "red", alpha=0.3)
```



Trun map into heat map to show density

```
chicago.homicides.coords <- chicago.homicides |>
  sf::st_coordinates() |>
  as.data.frame() |>
  dplyr::bind_cols(chicago.homicides)

ggplot() +
  ## Density heat map
  geom_density2d_filled(
    data = chicago.homicides.coords,
    aes(x = X, y = Y, fill = after_stat(level)),
    alpha = 0.85,
    contour_var = "ndensity"
  ) +

  scale_fill_viridis_d(
    option = "magma",
    direction = -1,
    name = "Density Level"
  ) +
  ## Chicago B0oundary
  geom_sf(data = chicago.boundary, color = "orange", size = 1, alpha = 0) +

  labs(
```

```

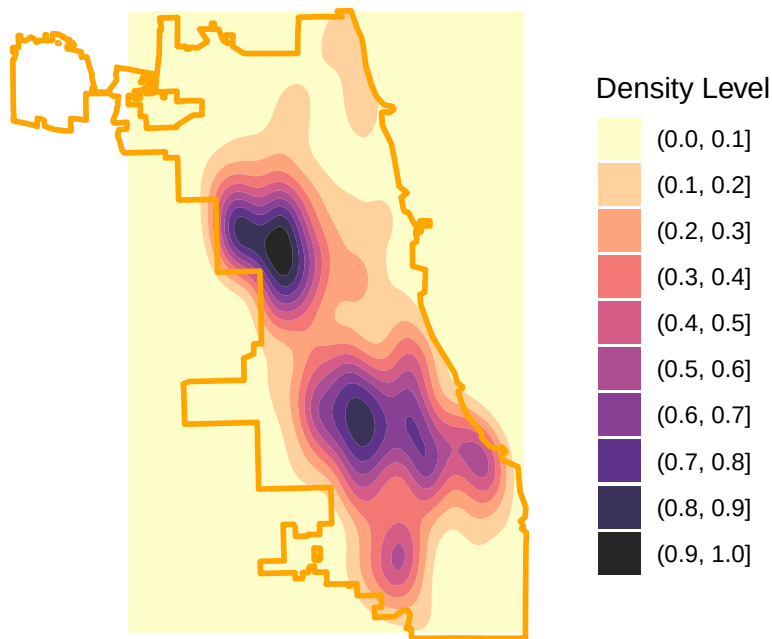
title = "Homicide Density Across Chicago",
subtitle = "Kernel density estimate of homicide locations",
caption = paste0("Total number of homicides: ", nrow(chicago.homicides))
) +

theme_minimal() +
theme(
  panel.grid = element_blank(),
  axis.text = element_blank(),
  axis.title = element_blank(),
  legend.position = "right",
  plot.title = element_text(face = "bold", size = 18),
  plot.subtitle = element_text(size = 13),
  plot.caption = element_text(size = 9, color = "grey40")
)

```

Homicide Density Across Chicago

Kernel density estimate of homicide locations



Total number of homicides: 5535

```
ggsave('Homicide_Density_Accross_Chicago.png')
```

Interpretation:

- Areas with highest homicide density in the area are in the darkest color contour
- Each lighter contour shows areas with proportionally lower relative density to the darkest contour

Solved/unsolved locations

```
chicago.homicides.top.3.races = chicago.homicides |>
  sf::st_drop_geometry() |>
  dplyr::group_by(victim_race) |>
  dplyr::summarise(count = dplyr::n()) |>
  dplyr::arrange(-count) |>
  dplyr::slice(1:3) |>
  dplyr::select(victim_race) # |>
  # as.vector()

chicago.homicides.top.3.races = chicago.homicides |>
  # sf::st_drop_geometry() |>
  dplyr::mutate(solved = ifelse(disposition %in% c("Closed by arrest"), "Solved", "Unsolved"))
  dplyr::filter(victim_race %in% chicago.homicides.top.3.races$victim_race)

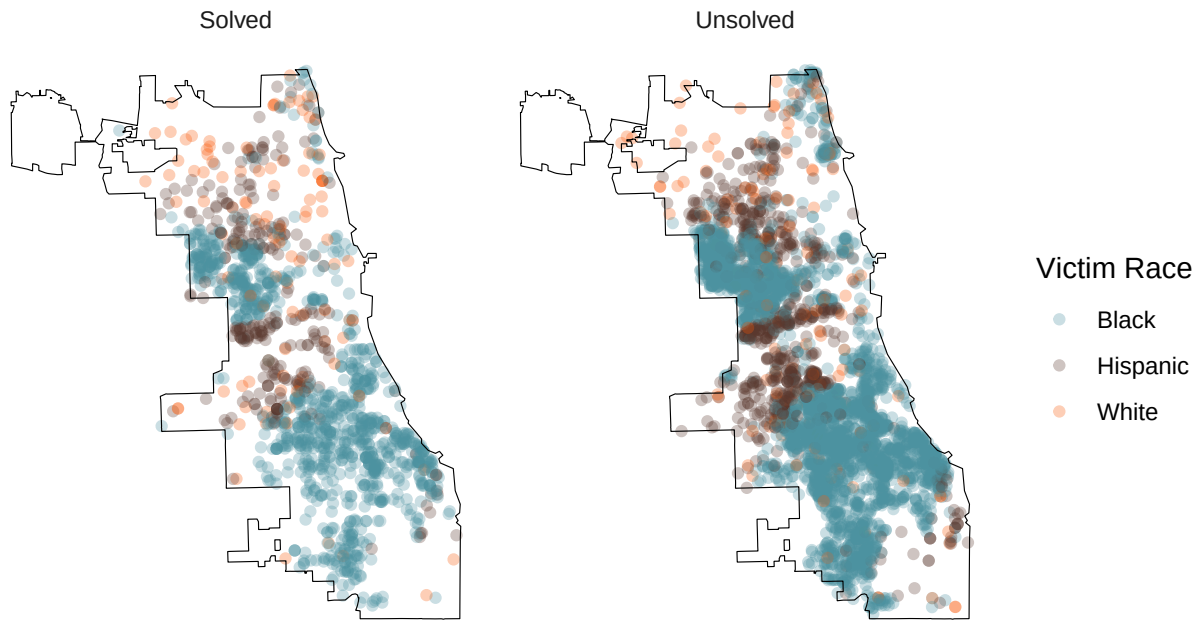
ggplot(data = chicago.homicides.top.3.races) +
  geom_sf(aes(colour = victim_race), alpha = 0.3) +
  scale_color_manual(values = c("#4c92a0", "#5a382e", "#fa5a09")) +
  geom_sf(data = chicago.boundary, color = "black", alpha = 0) +
  facet_wrap(vars(solved)) +

  labs(title = "Homicides in Chicago",
        subtitle = "Three race groups with the highest homicide rate",
        color = "Victim Race") +

  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.text = element_blank(),
    axis.title = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold", size = 18),
    plot.subtitle = element_text(size = 13),
    plot.caption = element_text(size = 9, color = "grey40")
  )
```

Homicides in Chicago

Three race groups with the highest homicide rate



```
ggsave('Homicide_By_Race_Across_Chicago.png')
```